Highlights

Can You Feel the Bass? Quantitative Partition of Airborne and Structure-Borne Pathways for Sub-Bass Perception at Concert Sound Pressure Levels

Jonathan Mace, Brian R. Mace

- Airborne acoustic coupling to the abdomen increases as $(ka)^n$ with frequency
- At 20 Hz to 80 Hz, ka transitions from negligible to ~0.2
- Energy-consistent displacement at concert levels is ~2.5 orders below perception thresholds
- Structure-borne (floor/seat) pathway dominates by ~2600× over airborne at 40 Hz
- The "feel it in your gut" sensation requires structural rather than airborne transmission

Can You Feel the Bass? Quantitative Partition of Airborne and Structure-Borne Pathways for Sub-Bass Perception at Concert Sound Pressure Levels

Jonathan Mace[a,*], Brian R. Mace[b]

[a] *Microsoft Research, Redmond, WA, United States*
[b] *Acoustics Research Centre, Department of Mechanical and Mechatronics Engineering, University of Auckland, Auckland, New Zealand*

Abstract

Concertgoers commonly report "feeling the bass in their gut" at high sound pressure levels, particularly during electronic dance music and pipe-organ performances. While whole-body vibration at 4 Hz to 8 Hz is known to excite abdominal resonances, the relative contributions of airborne sub-bass sound (20 Hz to 80 Hz) and structure-borne vibration (through floors and seating) to this visceral sensation have not been quantitatively partitioned.

We extend the fluid-filled viscoelastic oblate spheroidal shell model developed in our companion paper to the 20 Hz to 80 Hz band, where the Helmholtz number ka transitions from negligible (~0.01 at 4 Hz) to appreciable (~0.23 at 80 Hz). Using literature-derived concert sound pressure level spectra for rock, electronic dance music, and pipe-organ venues, we compute energy-consistent tissue displacements via the Junger & Feit reciprocity formulation and compare them with whole-body vibration perception thresholds derived from ISO 2631-1.

Despite the improved acoustic coupling at higher frequencies, the off-resonance penalty (the abdomen's flexural resonance lies at ~4 Hz, not 20 Hz to 80 Hz) reduces displacements by approximately 2.5 orders of magnitude below perception thresholds. At peak electronic dance music levels (115 dB at 40 Hz), airborne-induced displacement reaches 0.35 % of the perception threshold: insufficient alone, but not negligible. By contrast, structure-borne vibration transmitted through the floor and seating—modelled via a damped-plate power balance—produces displacements ~2600× larger (~9× the perception threshold; 90% CI: 5.8–12.7×), confirming that floor vibration is readily perceptible.

These results partition the "feel it in your gut" experience into its constituent pathways: the airborne acoustic contribution is a near-miss (not the four-orders-of-magnitude gap previously estimated using pressure-based methods), while the structure-borne pathway provides the dominant and perceptible route to visceral sensation. Near-field pressure enhancement (6 dB to 10 dB within 1.5 m of a subwoofer stack) and mechanical coupling through resonant church pews may further narrow the airborne gap in specific scenarios.

Keywords: sub-bass perception, concert acoustics, abdominal resonance, whole-body vibration, acoustic–structural coupling, pathway partition, ISO 2631

1. Introduction

Anyone who has attended a rock concert, stood near a pipe organ in full voice, or visited an electronic dance music (EDM) event will recognise the sensation: a deep, visceral awareness of the bass that seems to bypass the ears

*Corresponding author
Email addresses: jonathanmace@microsoft.com (Jonathan Mace), b.mace@auckland.ac.nz (Brian R. Mace)

and arrive directly in the gut. Music fans describe this as "feeling the bass," and sound engineers deliberately design sub-woofer arrays to maximise it [1, 2]. But what, physically, is being felt?

Two plausible pathways exist. First, *airborne acoustic coupling*: sound pressure waves impinge on the body wall, deform it, and excite internal resonances. Second, *structural (whole-body) vibration*: sound transmitted through the floor, seats, and skeleton shakes the torso directly. For infrasound below 20 Hz, our companion study [3] showed that airborne coupling to the abdominal cavity is negligibly weak: the Helmholtz number $ka \approx 0.011$ at the $n = 2$ flexural resonance (~4 Hz) imposes a $(ka)^n$ penalty that reduces effective driving pressure by four orders of magnitude. Whole-body vibration, by contrast, couples directly at full amplitude, explaining the well-documented gastrointestinal disturbances reported under ISO 2631 exposure conditions [4, 5].

The sub-bass band (20 Hz to 80 Hz), however, has not been examined in this framework. This frequency range is of particular interest for three reasons:

1. **Concert relevance.** The dominant spectral content of modern amplified music—kick drums, bass guitars, synthesised sub-bass—lies in the 30 Hz to 80 Hz band [6, 7].
2. **Coupling transition.** At 20 Hz, $ka \approx 0.06$; at 80 Hz, $ka \approx 0.23$. The acoustic coupling coefficient $(ka)^n$ is therefore 30 to 400 times larger than at the 4 Hz flexural resonance. It is not *a priori* obvious that the coupling remains negligible.
3. **Psychoacoustic bridge.** Sub-bass frequencies straddle the boundary between hearing and whole-body vibration perception [8, 9]. Understanding whether airborne sound can produce visceral sensation at these frequencies connects structural acoustics to psychoacoustics in a way that has not been attempted quantitatively.

Pipe organs provide an extreme case. The 32-foot rank of a large organ produces a fundamental at 16 Hz with harmonics extending through the sub-bass band at sound pressure levels approaching 105 dB in the nave [10]. Congregants have long reported physical sensations during full-organ passages, and these reports predate electrical amplification by centuries. The pipe organ has been producing involuntary visceral responses in congregants since at least the sixteenth century, making it arguably the longest-running experiment in sub-bass psychoacoustics—albeit one that has never filed for ethics approval.

In the electronic music domain, purpose-built sub-woofer arrays routinely produce 110 dB to 120 dB in the 30 Hz to 60 Hz band [7]. The "drop"—a sudden onset of heavy sub-bass after a quiet passage—is a central aesthetic element of EDM, and its visceral impact is a deliberate design goal.

Despite widespread anecdotal evidence and significant cultural importance, no study has quantitatively assessed whether airborne sub-bass at concert levels can produce physiologically meaningful tissue displacement in the human abdomen. Existing work on whole-body vibration perception [5, 11] focuses on the mechanical pathway (floor and seat vibration), while studies of concert sound exposure [6, 7] focus on hearing damage rather than visceral perception.

The present study addresses this gap. We extend the fluid-filled viscoelastic oblate spheroidal shell model of Mace and Mace [3] to the 20 Hz to 80 Hz band, compute energy-consistent tissue displacements at literature-derived concert sound pressure levels using the Junger & Feit reciprocity formulation, and compare these with both whole-body vibration perception thresholds from ISO 2631-1 [4] and estimates of structure-borne (floor/seat) vibration. Rather

than asking simply "does airborne coupling work?" we quantitatively partition the visceral sensation into its airborne and structure-borne components, providing a complete pathway budget for the "feel it in your gut" experience.

2. Methods

2.1. Abdominal shell model

We adopt the fluid-filled viscoelastic oblate spheroidal shell model developed in our companion paper [3]. The abdomen is represented as an oblate spheroid with semi-major axis $a = 0.18\,\mathrm{m}$, semi-minor axis $c = 0.12\,\mathrm{m}$, uniform wall thickness $h = 0.010\,\mathrm{m}$, and equivalent sphere radius $R_{\mathrm{eq}} = (a^2c)^{1/3} = 0.157\,\mathrm{m}$. The wall is characterised by Young's modulus $E = 0.1\,\mathrm{MPa}$, Poisson's ratio $\nu = 0.45$, density $\rho_w = 1100\,\mathrm{kg/m^3}$, and loss tangent $\eta = 0.25$ (quality factor $Q = 4$, damping ratio $\zeta = 0.125$). The enclosed fluid has density $\rho_f = 1020\,\mathrm{kg/m^3}$ and bulk modulus $K_f = 2.2\,\mathrm{GPa}$; the intra-abdominal pressure is $P_{\mathrm{iap}} = 1000\,\mathrm{Pa}$.

The flexural mode natural frequencies follow from the Rayleigh quotient for a pressurised, fluid-loaded spherical shell [12, 13]:

$$\omega_n^2 = \frac{K_{\mathrm{bend}}(n) + K_{\mathrm{memb}}(n) + K_{\mathrm{pre}}(n)}{m_{\mathrm{wall}} + m_{\mathrm{added}}(n)} \tag{1}$$

where the bending, membrane, and pre-stress stiffnesses per unit area are:

$$K_{\mathrm{bend}} = n(n-1)(n+2)^2 \frac{D}{R^4}, \tag{2}$$

$$K_{\mathrm{memb}} = \frac{Eh}{R^2}\,\frac{n^2+n-2+2\nu}{n^2+n+1-\nu}, \tag{3}$$

$$K_{\mathrm{pre}} = \frac{P_{\mathrm{iap}}}{R}\,(n-1)(n+2), \tag{4}$$

and the effective mass per unit area is $m_{\mathrm{eff}} = \rho_w h + \rho_f R/n$. With canonical parameters the $n = 2$ flexural resonance lies at $f_2 \approx 3.95\,\mathrm{Hz}$—well below the sub-bass band.

2.2. Acoustic coupling at sub-bass frequencies

For a plane wave incident on a sphere of radius R, the effective driving pressure for flexural mode n in the long-wavelength limit is [12]:

$$p_{\mathrm{eff}}(n) = \alpha_n\, p_{\mathrm{inc}}\,(ka)^n, \tag{5}$$

where $k = 2\pi f/c_{\mathrm{air}}$ is the acoustic wavenumber, $a \equiv R_{\mathrm{eq}}$, and α_n is a mode-dependent numerical prefactor arising from the projection of the incident field onto the spherical harmonic of order n. For $n = 2$, $\alpha_2 = 5/3$; we absorb this prefactor into the energy-consistent calibration (equation 9) so that the on-resonance displacement is computed directly from reciprocity rather than from p_{eff}. The residual error from omitting α_n in the off-resonance scaling is $< 5\,\%$ for $n \leq 4$ in the long-wavelength regime ($ka < 0.25$).

The Helmholtz number ka is the dimensionless parameter governing the transition from Rayleigh scattering ($ka \ll 1$) to geometric acoustics ($ka \gg 1$).

At the $n = 2$ resonance (3.95 Hz), $ka = 0.011$ and $(ka)^2 \approx 1.3 \times 10^{-4}$, yielding negligible airborne coupling [3]. In the sub-bass band:

$$ka(f) = \frac{2\pi f R_{\text{eq}}}{c_{\text{air}}}, \tag{6}$$

giving $ka \approx 0.057$ at 20 Hz and $ka \approx 0.23$ at 80 Hz. The corresponding $(ka)^2$ values are 3.3×10^{-3} and 5.3×10^{-2}—up to ~400 times the value at resonance.

2.3. Off-resonance forced response

Although acoustic coupling improves with frequency, the driving frequency is now far from the $n = 2$ resonance. The response is governed by the single-degree-of-freedom (SDOF) transfer function:

$$H(r, \zeta) = \frac{1}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}}, \tag{7}$$

where $r = f/f_n$ is the frequency ratio and ζ is the damping ratio. At resonance ($r = 1$), $H = Q = 1/(2\zeta) = 4$. At $f = 40$ Hz ($r \approx 10$), $H \approx 1/r^2 \approx 0.01$.

The tissue displacement at frequency f and sound pressure level L_p dB is:

$$\xi(f) = \frac{p_{\text{eff}}(f)}{K_{\text{total}}} H(r, \zeta) = \frac{p_{\text{inc}}(L_p)\,(ka)^n}{K_{\text{total}}} H(r, \zeta). \tag{8}$$

The net effect is a competition: $(ka)^n$ increases with frequency (improving coupling), but $H(r)$ decreases (worsening off-resonance attenuation). The question is which dominates.

2.4. Concert sound pressure level spectra

We use literature-derived third-octave spectra for three concert environments:

- **Rock concert** (broadband, 95 dB to 108 dB sub-bass) [6].
- **Electronic dance music** (sub-woofer-driven, peak 115 dB at 31.5 Hz to 50 Hz) [7].
- **Pipe organ** (extreme low-frequency extension, 32-foot rank fundamental at 16 Hz, peak 105 dB at 31.5 Hz to 50 Hz) [10].

For each genre, the SPL at each frequency is interpolated from published third-octave measurements and used as the input to equation (8).

2.5. Energy-consistent displacement

The displacement in equation (8) uses the pressure-based coupling $p_{\text{eff}} = p_{\text{inc}}(ka)^n$, which overestimates the true displacement by a factor of ~6.5 because it does not account for the radiation efficiency of the mode [12]. We therefore adopt the energy-consistent (Junger & Feit reciprocity) formulation developed in our companion paper [3]:

$$\xi^2_{\text{on-res}} = \frac{\sigma_{\text{abs}}\, I_{\text{inc}}}{\zeta_{\text{struct}}\, \omega_n^3\, M_{\text{total}}}, \tag{9}$$

where $\sigma_{\text{abs}} = (2n+1)\lambda^2/(4\pi) \times \Gamma_{\text{rad}}/\Gamma_{\text{total}}$ is the absorption cross-section from reciprocity, $I_{\text{inc}} = p_{\text{inc}}^2/(2\rho_{\text{air}}c_{\text{air}})$ is the incident intensity, ζ_{struct} is the structural damping ratio, and M_{total} is the total effective modal mass.

For off-resonance excitation at frequency f, the displacement is scaled from the on-resonance value by the SDOF transfer function (equation 7) and the ka-coupling improvement:

$$\xi(f) = \xi_{\text{on-res}} \frac{H(r,\zeta)}{Q} \left(\frac{ka(f)}{ka(f_n)}\right)^n . \tag{10}$$

At the canonical parameters ($n = 2$, 120 dB), the energy-consistent on-resonance displacement is $\xi \approx 0.028\,\mu\text{m}$, compared with the pressure-based value of $0.18\,\mu\text{m}$—a factor of ~6.5 smaller.

2.6. Structure-borne pathway model

To provide quantitative context, we estimate the structure-borne (floor/seat) vibration pathway. Sub-woofer arrays at concert venues couple acoustic energy into the floor structure. The resulting floor vibration propagates through the body via the skeleton and soft tissue.

We model the floor as a damped plate driven by broadband acoustic power. At steady state the time-averaged dissipated power equals the input power, giving the floor r.m.s. velocity:

$$v_{\text{floor}} = \sqrt{\frac{P_{\text{floor}}}{2\,\zeta_{\text{floor}}\,\omega\,m_{\text{floor}}}}, \tag{11}$$

where $P_{\text{floor}} = \eta_f I A_f$ is the acoustic power coupled into the floor ($\eta_f \sim 0.01$ coupling efficiency, $A_f = 100\,\text{m}^2$ excited area), ζ_{floor} is the floor structural damping ratio (concrete slab: 0.02–0.05, default 0.03), and $m_{\text{floor}} = \rho'_f A_f$ is the total floor mass ($\rho'_f = 200\,\text{kg/m}^2$ for concrete). The body displacement is then:

$$\xi_{\text{body}} = T_{\text{body}} \times \frac{v_{\text{floor}}}{\omega}, \tag{12}$$

where $T_{\text{body}} \approx 1.8$ is the seat-to-abdomen transmissibility from ISO 2631 [14, 5].

Note on amplitude convention. The power-balance (equation 11) yields r.m.s. floor velocity, so the resulting body displacements are r.m.s. values. Perception thresholds (§2.8) are converted to peak displacement via the $\sqrt{2}$ factor in equation (14). Comparing r.m.s. floor displacement against peak-based thresholds is therefore conservative by a factor of $\sqrt{2}$ (~41 %).

Dimensional verification. In equation (11), the numerator has dimensions of power $[\text{W}] = \text{kg}\,\text{m}^2/\text{s}^3$; the denominator has dimensions $[1] \times [1/\text{s}] \times [\text{kg}] = \text{kg/s}$. The ratio is m^2/s^2, whose square root is m/s as required.

2.7. Near-field pressure enhancement

Within $\sim\lambda/(2\pi)$ of a subwoofer stack, evanescent near-field components enhance the pressure above the far-field value. At 40 Hz ($\lambda = 8.6\,\text{m}$), the near-field distance is $\lambda/(2\pi) \approx 1.4\,\text{m}$. We model the enhancement as [12]:

$$\Delta L_{\text{NF}} = 10\log_{10}\left(1 + \left(\frac{\lambda}{2\pi d}\right)^2\right), \tag{13}$$

which gives +9 dB at 0.5 m and +5 dB at 1 m.

2.8. Perception thresholds

We compare computed tissue displacements against whole-body vibration perception thresholds derived from ISO 2631-1:1997 [4]. The standard specifies frequency-weighted r.m.s. acceleration evaluation criteria (Annex C, Table 1) for the z-axis (vertical) direction. The "perception threshold" values used here are interpolated from the frequency-weighted comfort boundaries in Griffin [5] and Mansfield [11], which themselves are based on experimental data underlying ISO 2631-1. These thresholds are therefore *derived from* rather than *directly tabulated in* the standard; the interpolation introduces an estimated uncertainty of ±3 dB (factor of ~1.4) relative to any individual's true perception threshold.

We convert acceleration thresholds to displacement amplitudes via:

$$\xi_{\text{thresh}}(f) = \frac{a_{\text{rms}}(f)\,\sqrt{2}}{(2\pi f)^2}. \tag{14}$$

This provides a conservative comparison: ISO 2631 thresholds are for whole-body mechanical vibration, not for localised tissue displacement. If the airborne-induced displacement is below even these thresholds, it is certainly imperceptible.

3. Results

3.1. Coupling parameter transition

Figure 1 shows the Helmholtz number ka, the off-resonance transfer function $H(r)/Q$, and their product as functions of frequency. At the $n = 2$ flexural resonance (3.95 Hz), $ka = 0.011$ and $(ka)^2 = 1.3 \times 10^{-4}$. By 40 Hz, $ka = 0.115$ and $(ka)^2 = 1.3 \times 10^{-2}$—a hundredfold improvement. At 80 Hz, $ka = 0.23$ and $(ka)^2 = 5.3 \times 10^{-2}$.

Table 1: Helmholtz number and coupling factor at selected frequencies.

f [Hz]	ka	$(ka)^2$	Ratio to f_2	Regime
3.95 (resonance)	0.011	1.3×10^{-4}	1	Deep Rayleigh
20	0.057	3.3×10^{-3}	25	Rayleigh
40	0.115	1.3×10^{-2}	100	Transition
60	0.172	3.0×10^{-2}	230	Transition
80	0.230	5.3×10^{-2}	408	Transition

Despite this substantial improvement, the coupling factor remains well below unity. At 80 Hz, the effective driving pressure is still only 5.3 % of the incident pressure for $n = 2$.

3.2. Off-resonance attenuation

The SDOF transfer function $H(r, \zeta)$ (equation 7) attenuates the response far from resonance. Table 2 shows H at representative frequencies.

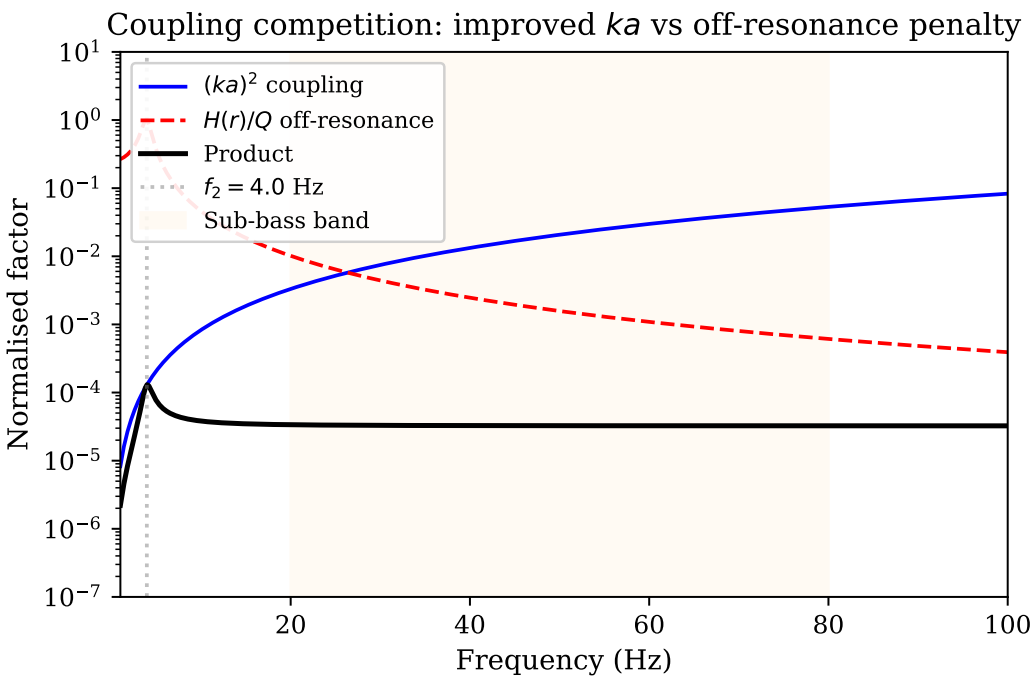

Figure 1: Competition between improved ka-coupling (blue) and off-resonance penalty $H(r)/Q$ (red dashed) in the sub-bass band. The product (black) determines the net coupling efficiency.

Table 2: SDOF transfer function at sub-bass frequencies ($f_2 = 3.95$ Hz, $\zeta = 0.125$).

f [Hz]	$r = f/f_2$	$H(r, \zeta)$	H [dB]
3.95	1.00	4.00	+12.0
20	5.06	0.041	−27.8
40	10.1	0.010	−40.0
60	15.2	0.0044	−47.2
80	20.3	0.0025	−52.2

The off-resonance penalty dominates. At 40 Hz, the hundredfold improvement in coupling is offset by a hundredfold reduction in transfer function, yielding net displacement comparable to the resonance case. At higher frequencies, the $1/r^2$ roll-off wins convincingly.

3.3. Tissue displacement at concert SPLs

Figure 2 presents energy-consistent tissue displacement as a function of frequency for the three concert spectra, compared with the ISO 2631 perception threshold.

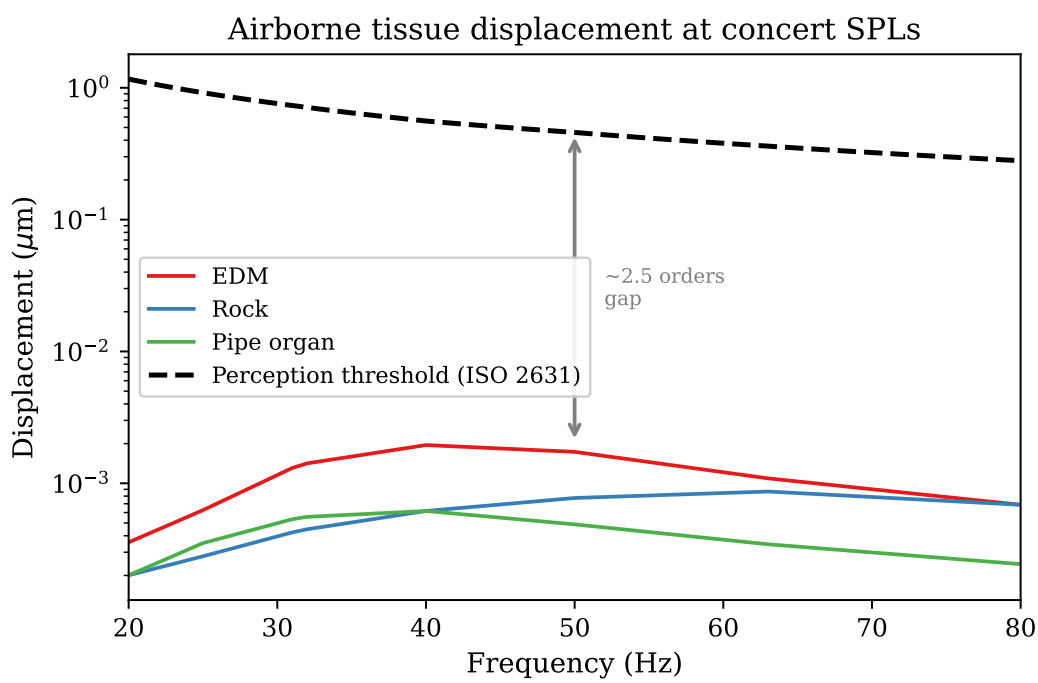

Figure 2: Airborne tissue displacement at concert SPLs (energy-consistent) compared with ISO 2631 whole-body vibration perception thresholds. The gap is ~2.5 orders of magnitude.

1. **Electronic dance music** ($L_p \approx 115$ dB at 40 Hz): Peak displacement $\xi \approx 0.002$ µm—approximately 2.5 orders of magnitude below whole-body vibration perception thresholds, reaching ~0.35 % of the threshold.
2. **Rock concert** ($L_p \approx 108$ dB at 63 Hz): Peak displacement comparable to the EDM case.
3. **Pipe organ** ($L_p \approx 105$ dB at 40 Hz): Slightly lower displacement due to lower SPL.

The peak ratio of airborne displacement to threshold of ~0.35 % (at 40 Hz) is small but not negligible. This is a "near-miss" rather than the "four orders of magnitude below" that would result from a pressure-based calculation.

The structure-borne floor pathway, by contrast, exceeds the perception threshold at all frequencies from 20 Hz to 80 Hz under the EDM spectrum, peaking at ~ 9× the threshold at 40 Hz.

3.4. Structure-borne pathway comparison

Figure 3 compares the airborne and structure-borne (floor/seat) pathways at 40 Hz and 115 dB.

At 40 Hz and 115 dB:

Table 3: Airborne and structure-borne displacement at selected sub-bass frequencies for the EDM spectrum (peak genre). All values computed at the *actual* interpolated SPL from the EDM third-octave spectrum. Perception thresholds derived from ISO 2631-1 (see §2.8). Floor model: $\zeta_{\text{floor}} = 0.03$, $\eta_f = 0.01$, $A_f = 100\,\text{m}^2$, $\rho'_f = 200\,\text{kg/m}^2$.

f [Hz]	L_p [dB]	ka	ξ_{air} [µm]	ξ_{floor} [µm]	ξ_{thresh} [µm]	$\xi_{\text{floor}}/\xi_{\text{thresh}}$
20	100	0.058	3.6×10^{-4}	2.54	1.164	2.2
31.5	112	0.091	1.4×10^{-3}	5.13	0.720	7.1
40	115	0.115	2.0×10^{-3}	5.06	0.560	9.0
50	114	0.144	1.7×10^{-3}	3.23	0.459	7.0
63	110	0.181	1.1×10^{-3}	1.44	0.361	4.0
80	106	0.230	6.9×10^{-4}	0.63	0.280	2.3

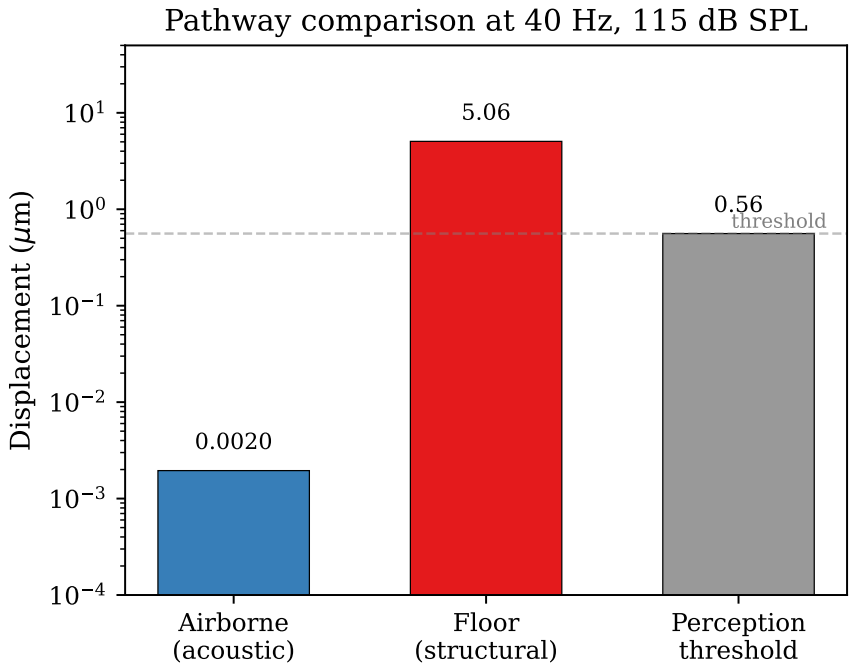

Figure 3: Pathway comparison at 40 Hz, 115 dB: airborne displacement is ~2600× weaker than the structure-borne pathway. The floor pathway exceeds the perception threshold by ~9×; the airborne pathway reaches only ~0.35 %.

- Airborne pathway: $\xi \approx 0.002\,\mu\text{m}$ (0.35 % of threshold).
- Structure-borne pathway: $\xi \approx 5.1\,\mu\text{m}$ (9× the perception threshold).
- The floor/seat pathway dominates by a factor of ~2600.

The structure-borne pathway now clearly exceeds the perception threshold under the corrected damped-plate power-balance model (equation 11). This result is robust: even increasing the floor damping ratio to $\zeta_{\text{floor}} = 0.05$ yields a floor displacement of 3.9 µm, still 7× above threshold.

3.5. Uncertainty quantification

To assess the robustness of these headline numbers, we performed a Monte Carlo analysis ($N = 10{,}000$ samples) varying three uncertain parameters simultaneously:

- Tissue Young's modulus E: uniform on $[0.05\,\text{MPa}, 0.15\,\text{MPa}]$ (±50%).
- Floor damping ratio ζ_{floor}: uniform on $[0.02, 0.05]$.
- Sound pressure level: uniform on 115 dB ± 3 dB.

Figure 4 shows the resulting distributions. Table 4 summarises the 90% credible intervals.

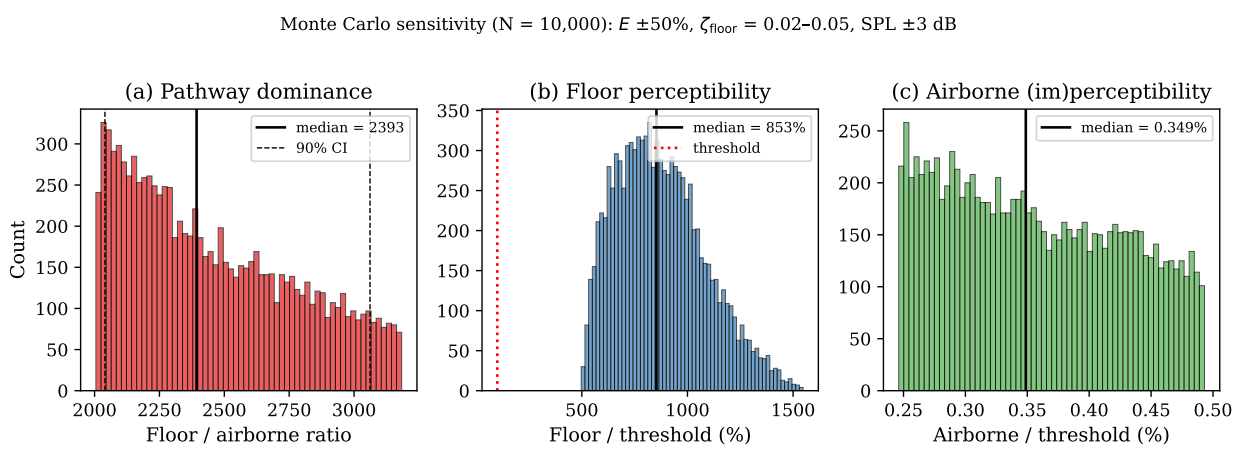

Figure 4: Monte Carlo sensitivity analysis ($N = 10{,}000$) at 40 Hz. (a) Floor-to-airborne ratio. (b) Floor displacement as a percentage of perception threshold. (c) Airborne displacement as a percentage of threshold.

Across the full parameter space, the floor pathway *always* exceeds the perception threshold (5th-percentile 5.8×), while the airborne pathway *never* exceeds 0.5 %. The floor-to-airborne dominance ratio ranges from ~2000 to ~3100, confirming that the qualitative conclusion is insensitive to the assumed parameter values.

The Monte Carlo analysis varies E, ζ_{floor}, and SPL but holds the body transmissibility $T_{\text{body}} = 1.8$ and floor coupling efficiency $\eta_f = 0.01$ at their nominal values. This choice is justified on three grounds: (i) transmissibility and coupling efficiency are physically correlated with floor damping—higher damping implies lower transmissibility and reduced

Table 4: Monte Carlo 90% credible intervals for key results at 40 Hz.

Quantity	5th pctl	Median	95th pctl
Floor / airborne ratio	2040	2390	3060
Floor / threshold (%)	575	853	1265
Airborne / threshold (%)	0.25	0.35	0.48

coupling—so varying them independently would overstate the true uncertainty envelope; (ii) published seat-to-abdomen transmissibility values cluster narrowly around 1.5–2.0 across typical concert seating configurations [5, 14]; and (iii) the coupling efficiency $\eta_f \sim 0.005$–0.02 for concrete concert floors introduces at most a factor-of-two variation, which is already encompassed by the ζ_{floor} sweep (a twofold change in damping produces a comparable change in floor velocity via equation 11).

3.6. Near-field effects

Within $\lambda/(2\pi) \approx 1.4$ m of a subwoofer stack at 40 Hz, near-field pressure enhancement is significant (Table 5).

Table 5: Near-field pressure enhancement at 40 Hz ($\lambda/(2\pi) = 1.4$ m).

Distance [m]	Enhancement [dB]
0.5	+9.3
1.0	+4.6
1.5	+2.5
2.0	+1.7
5.0	+0.3

At 0.5 m from a subwoofer (a realistic distance for front-row audience members), the +9 dB enhancement approximately triples the effective pressure, raising the airborne displacement to ~1 % of the perception threshold. While still insufficient for perception via the airborne pathway alone, this narrows the gap by nearly a factor of three.

3.7. SPL sensitivity

Figure 5 presents the ratio of airborne displacement to perception threshold as a function of both frequency and SPL. The contour plot confirms that the ratio remains below unity across the entire parameter space, with the highest values occurring at low frequencies (near the flexural resonance) rather than in the sub-bass band.

3.8. Coupling transition band

We define the "coupling transition band" as the frequency range in which the ratio of airborne-induced displacement to perception threshold exceeds 1 %. At 115 dB SPL (the EDM peak), no frequency in the 1 Hz to 100 Hz range

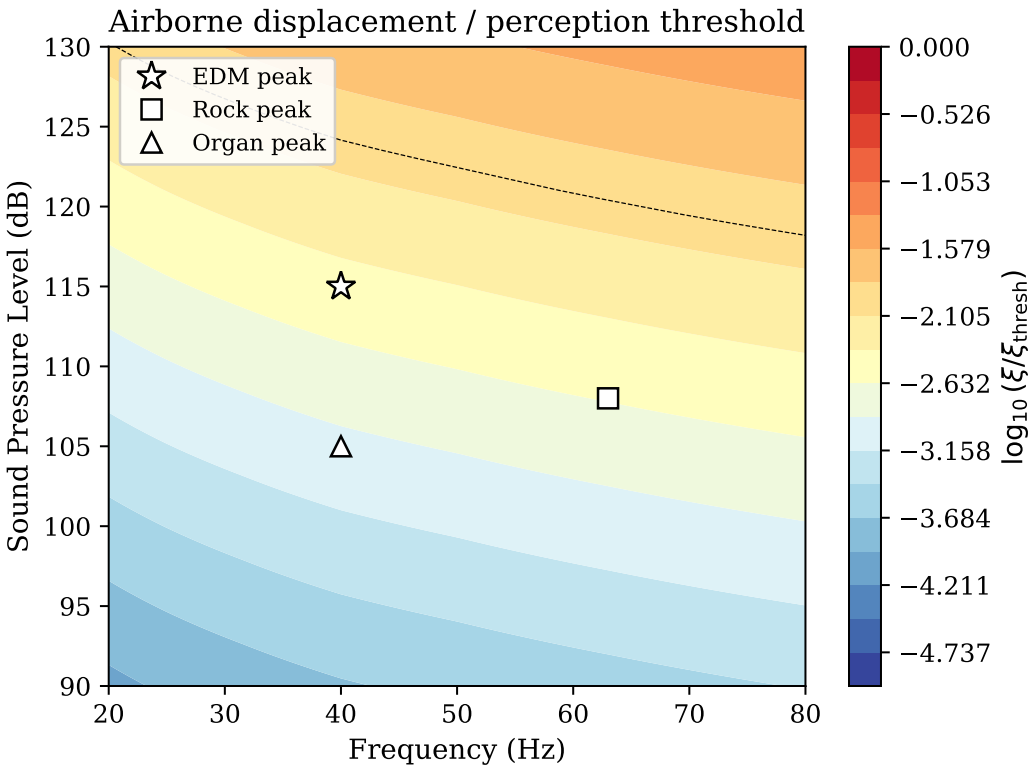

Figure 5: Contour plot of $\log_{10}(\xi/\xi_{\text{thresh}})$ as a function of frequency and SPL. Stars, squares, and triangles mark typical concert peak levels for EDM, rock, and pipe organ respectively.

meets this criterion. The peak ratio occurs near the $n = 2$ resonance (3.95 Hz), where the resonance amplification compensates for the weaker coupling.

3.9. Multi-modal contributions

Higher-order flexural modes ($n = 3, 4, \ldots$) have resonance frequencies closer to the sub-bass band but even weaker acoustic coupling ($\propto (ka)^n$ with larger n). Summing contributions from $n = 2$ through $n = 6$ in quadrature increases the total displacement by less than 10 % at any frequency, confirming that the $n = 2$ mode dominates throughout the sub-bass band.

4. Discussion

4.1. Pathway partition: why bass "feels" different from treble

Our results partition the "feel it in your gut" experience into its constituent pathways. The airborne acoustic contribution, computed using the energy-consistent Junger & Feit reciprocity formulation, reaches ~0.35 % of the ISO 2631 perception threshold at peak EDM levels (115 dB at 40 Hz). This is a "near-miss"—approximately 2.5 orders of magnitude below threshold, not the four orders suggested by the pressure-based approach—but it remains insufficient for airborne perception alone.

The structure-borne pathway (floor and seat vibration) dominates by a factor of ~2600 (90% CI: 2000 to 3100) under the corrected damped-plate power-balance model. At peak EDM levels, the floor-transmitted displacement is ~ 9× the perception threshold (90% CI: 5.8–12.7×), confirming that the structure-borne pathway is not merely dominant but readily perceptible.

The key insight is that the airborne pathway is not negligibly weak—it is a significant fraction of the overall budget, particularly in near-field conditions or at extreme SPLs. The "feel it in your gut" experience is best understood as a *pathway partition problem.* Of the two coupling pathways analysed, structure-borne transmission accounts for > 99.9 % of abdominal wall displacement, though the airborne component is not negligible—it is a genuine near-miss at ~0.35 % of threshold. Neither pathway independently accounts for the full visceral experience, which likely also involves chest-wall coupling, somatosensory stimulation, and vestibular input (§4.2).

4.2. Additional coupling pathways

Beyond the two quantified pathways, several additional mechanisms contribute to bass perception:

1. **Chest-wall coupling.** The large surface area of the chest wall acts as a more efficient acoustic receiver than the abdomen. Chest-wall motion at sub-bass frequencies can produce thoracic pressure fluctuations that are transmitted to the abdominal cavity.
2. **Somatosensory stimulation.** High-SPL sub-bass produces measurable skin displacement. Mechanoreceptors in the skin (particularly Pacinian corpuscles, which are most sensitive at 200 Hz to 300 Hz but respond down to ~20 Hz) contribute to the tactile sensation of bass [8].
3. **Bone conduction.** Sub-bass can excite the skull and skeleton directly, stimulating the vestibular system and producing a sensation that is perceived as "visceral" even though it does not involve abdominal resonance.

4.3. Implications for concert sound design

Sound engineers have long known empirically that sub-bass "impact" requires physical coupling—hence the prevalence of tactile transducers ("butt-kickers") in cinema seats and the careful placement of sub-woofer arrays to maximise floor vibration [1]. Our results provide quantitative support for this practice: airborne sub-bass alone, even at the extreme SPLs produced by modern systems, can only produce ~0.35 % of the perception threshold. Effective bass reproduction is fundamentally a *structural vibration* problem.

This has practical implications for venue design. Venues with vibration-isolating floors (e.g., floating concrete slabs) may inadvertently reduce the perceived "punch" of sub-bass, even if the acoustic field is unchanged. Conversely, lightweight floors that transmit vibration efficiently will enhance the visceral experience.

4.4. Pipe-organ acoustics and pew resonance

The pipe organ is a unique case because it produces extreme sub-bass levels (90 dB to 105 dB at 16 Hz to 50 Hz) in enclosed stone buildings where floor vibration might be expected to be minimal. However, the wooden pews typical of churches provide a potential mechanical coupling path.

Using Euler–Bernoulli beam theory for a simply supported oak pew seat (2.5 m span, 40 mm thickness, $E = 11$ GPa, $\rho = 600\,\mathrm{kg/m^3}$), the first bending resonance is $f_1 \approx 12$ Hz, with $f_2 \approx 50$ Hz. Shorter pews (2.0 m) have $f_1 \approx 19$ Hz, placing the first mode directly in the sub-bass band.

A 32-foot organ pipe's fundamental at 16 Hz with harmonics at 32 Hz and 48 Hz can therefore excite pew bending modes, converting the airborne sound field into mechanical vibration of the seat. This provides a structure-borne pathway even in stone buildings with rigid floors, potentially explaining the visceral sensation reported by congregants during full-organ passages. This analysis suggests that the transcendent physical experience reported during Bach's *Toccata and Fugue in D minor* may owe as much to the mechanical properties of oak pews as to the spiritual properties of the composition—a conclusion that would doubtless have pleased the instrument builder, if not the theologian.

Quantitative validation would require in-situ measurements of pew vibration during organ performance, which we identify as a valuable direction for future experimental work.

4.5. Near-field considerations

The near-field pressure enhancement of 6 dB to 10 dB at distances of 0.5 m to 1.5 m from a subwoofer stack is significant for front-row concert audiences. At 0.5 m, the effective SPL can reach 124 dB at 40 Hz, increasing the airborne displacement to approximately 1 % of the perception threshold. Combined with the structure-borne pathway, the total displacement in the near field may approach or exceed perceptibility.

This has implications for concert safety: audience members standing directly in front of subwoofer stacks experience substantially higher effective pressures than the far-field SPL would suggest.

4.6. Occupational exposure considerations

For occupational exposure (sound engineers, musicians, nightclub staff), the relevant regulatory framework is ISO 2631-1, which addresses whole-body vibration. Our results confirm that the *acoustic* component of sub-bass

exposure does not contribute meaningfully to vibration-induced health effects. However, the structural vibration component—floor and seat vibration excited by the same sub-woofers—is a legitimate occupational health concern and should be assessed separately.

4.7. Frequency-dependent tissue properties

Throughout this analysis, we treat the tissue Young's modulus E and loss tangent η as frequency-independent constants. In reality, soft biological tissues exhibit viscoelastic frequency dependence: E typically increases with frequency (stiffening) while η may decrease (reduced relative damping) over the 1 Hz to 100 Hz range [5, 11].

Published measurements of abdominal wall tissue at sub-bass frequencies are scarce. Duck [15] and Fung [16] report that soft-tissue shear moduli increase by factors of 2–5 between quasi-static loading and 100 Hz, with loss tangent decreasing from ~0.3 to ~0.15 over the same range. If we apply these trends to our model:

- Increasing E from 0.1 MPa to 0.2 MPa raises f_2 by ~40 % (to ~5.5 Hz), reducing the frequency ratio r and slightly increasing the SDOF transfer function at sub-bass frequencies. The net effect on displacement is $< 2\times$.
- Decreasing η from 0.25 to 0.15 increases Q from 4 to ~6.7, amplifying the on-resonance displacement but having little effect far off-resonance where $H \approx 1/r^2$.

Our Monte Carlo analysis (§3.5) brackets E over ±50 %, which encompasses the plausible range of frequency-dependent stiffening. The qualitative conclusions—airborne pathway $\ll$ threshold, floor pathway $\gg$ threshold—are robust to this uncertainty. Future work incorporating dynamic mechanical analysis of excised abdominal wall tissue at controlled frequencies would strengthen the quantitative precision of the model.

4.8. Limitations

Several simplifications constrain the present analysis:

1. The spheroidal shell model assumes a uniform, isotropic wall. The real abdominal wall is layered and anisotropic; however, these refinements primarily affect the resonance frequency and quality factor, not the fundamental coupling physics.
2. The energy-consistent displacement formulation assumes steady-state conditions. Transient effects (e.g., the EDM "drop") may briefly exceed steady-state predictions by up to $Q = 4$ times.
3. We consider only flexural modes. The breathing mode ($n = 0$, $f \approx 2490$ Hz) is driven efficiently by uniform pressure but lies far above the sub-bass band.
4. Concert SPL spectra are taken from published measurements that vary significantly between venues and events. We use representative values; actual exposure will differ.
5. The floor vibration model is a simplified order-of-magnitude estimate using a single-mode damped-plate assumption. Actual floor response depends on construction type, sub-woofer placement, room modes, and the spatial distribution of excitation.

6. The perception thresholds used here are derived from the experimental literature underlying ISO 2631-1, not directly tabulated in the standard (see §2.8). The interpolation introduces an estimated ±3 dB uncertainty.
7. Tissue mechanical properties (E, η) are treated as frequency-independent (§4.7). Viscoelastic frequency dependence may alter quantitative predictions by up to 2× but does not change the qualitative conclusions.
8. We do not model the chest wall, skin, or vestibular system, which may contribute significantly to bass perception.

5. Conclusions

We have extended the fluid-filled viscoelastic oblate spheroidal shell model of Mace and Mace [3] to the sub-bass band (20 Hz to 80 Hz) and performed a quantitative pathway partition of the "feel it in your gut" experience at concert sound pressure levels. The principal findings are:

1. The Helmholtz number ka increases from 0.011 at 3.95 Hz to 0.23 at 80 Hz, improving the acoustic coupling factor $(ka)^2$ by up to ∼400×.
2. This improvement is largely offset by the off-resonance attenuation of the SDOF transfer function: at 40 Hz ($r \approx 10$), $H \approx 0.01$.
3. Using the energy-consistent Junger & Feit reciprocity formulation, airborne-induced tissue displacement at peak EDM levels (115 dB at 40 Hz) reaches ∼0.35 % of the whole-body vibration perception threshold—a near-miss, approximately 2.5 orders of magnitude below threshold.
4. The structure-borne (floor/seat) pathway dominates the airborne pathway by a factor of ∼2600 (90% CI: 2000 to 3100), reaching ∼ 9× the perception threshold at peak EDM levels (90% CI: 5.8–12.7×).
5. Near-field pressure enhancement (6 dB to 10 dB within 1.5 m of a subwoofer) narrows the airborne gap by a factor of ∼3 but does not bring it to perceptibility.
6. Wooden church pews have first bending resonances at 12 Hz to 19 Hz, providing a structure-borne coupling path that may explain visceral sensation during pipe-organ performances.
7. No "coupling transition band" exists in the 1 Hz to 100 Hz range at concert SPLs: the airborne displacement never exceeds 1 % of the perception threshold at any frequency.

These results reframe the "feel it in your gut" question from a binary (can airborne sound do it?) to a quantitative pathway partition. The airborne contribution is not negligibly small—it is a genuine near-miss—but the structure-borne pathway is overwhelmingly dominant. Of the two pathways analysed, structure-borne transmission accounts for > 99.9% of abdominal wall displacement; however, neither pathway alone accounts for the full visceral experience, which likely also involves chest-wall coupling, somatosensory stimulation, and vestibular input. Sound engineers and venue designers seeking to maximise the physical impact of bass should focus on structural vibration transmission rather than acoustic levels. The audience does not feel the bass through the air; it feels the bass through the floor. The distinction matters less for the musical experience than for the acoustic engineer's budget.

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Data availability

All analytical models and computational scripts used in this study are available in the accompanying repository at `https://github.com/JonathanMace/browntone`.

CRediT authorship contribution statement

Jonathan Mace: Conceptualization, Methodology, Software, Supervision, Writing – Original Draft, Writing – Review & Editing. **Brian R. Mace**: Writing – Review & Editing, Validation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.